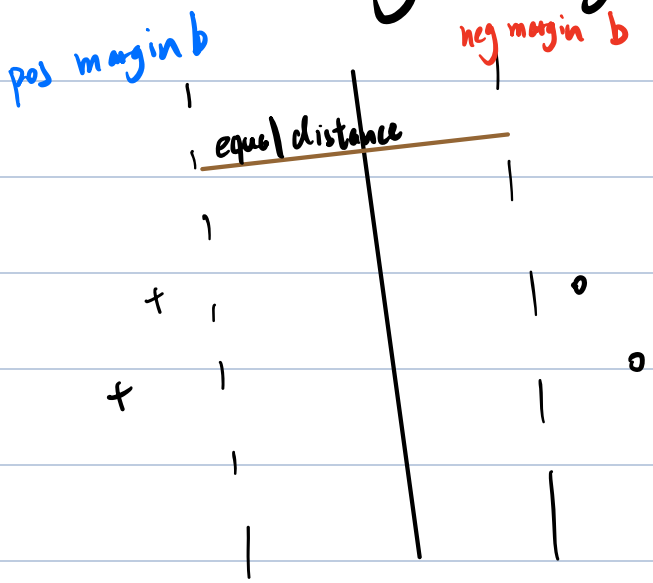
Large Margin Classifier: leave lots of space before hitting the training examples. (better)

Find large Margin Classifier



$$\theta \cdot x + \theta_0 = 0$$

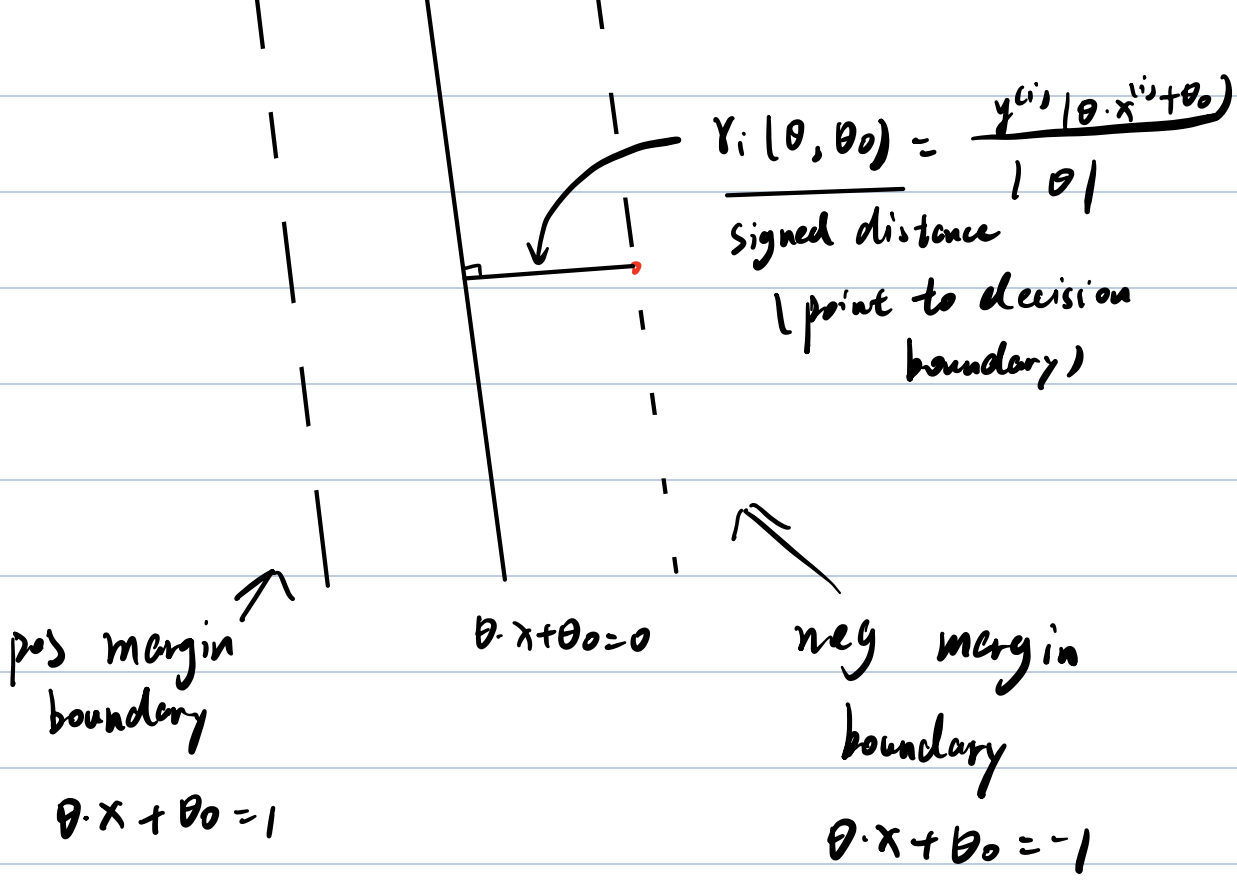
$$\Rightarrow \frac{\theta}{|\theta|} \cdot x + \frac{\theta_0}{|\theta|} = 0$$

The norm of θ controls the distance of margin boundary and the decision boundary.

decision boundary: $\theta \cdot x + \theta_0 = 0$

Margin boundary: $\theta \cdot x + \theta_0 = \pm 1$

Distance: $\frac{1}{|\theta|}$



Regularization goal: Maximize the distance between margin boundary and decision boundary.

Hinge Loss:

$$\text{Loss}_h\big(\overbrace{y^{(i)}(\theta \cdot x^{(i)} + \theta_0)}^{\text{agreement}(z)}\big) = \begin{cases} 0 & \text{if } z \geq 1 \\ 1 - z & \text{if } z < 1 \end{cases}$$

Regularization: $\max \frac{1}{|\theta|} \Rightarrow \min |\theta|$

The objective function

regularization parameter

$\lambda > 0$

$$J(\theta, \theta_0) = \frac{1}{n} \sum_{i=1}^{n} \text{Loss}\big(y^{(i)} \cdot (\theta \cdot x^{(i)} + \theta_0)\big) + \frac{\lambda}{2} |\theta|^2$$

$\Downarrow$

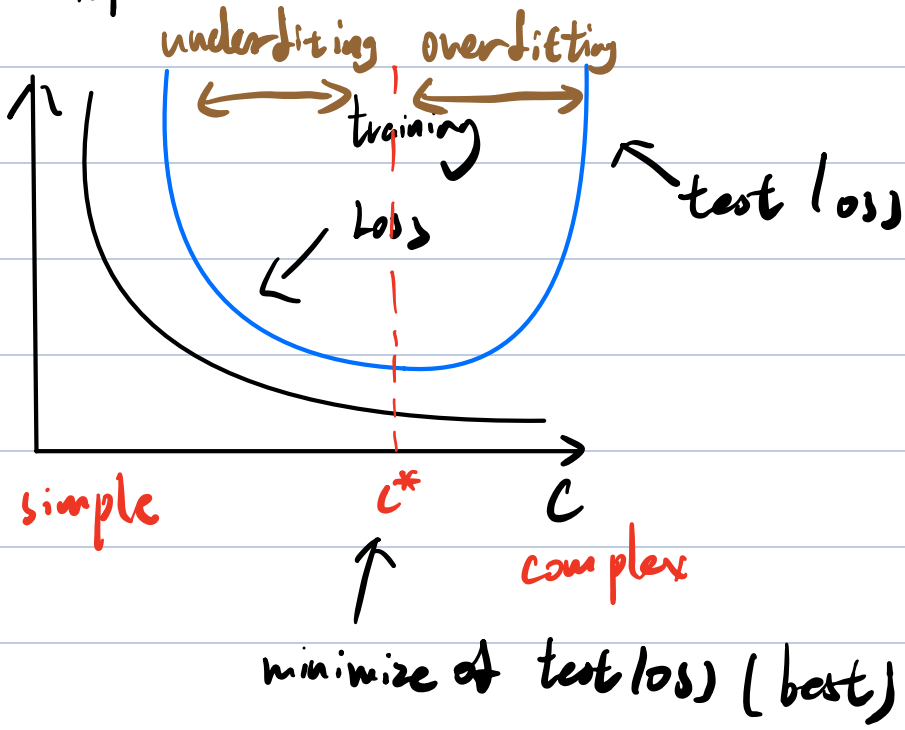
minimize $J(\theta, \theta_0)$

For Large λ: favor large margin

For small λ: favor putting the examples outside the margin boundary.

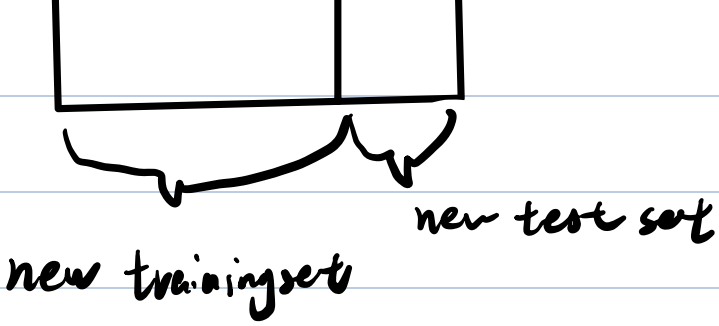
$$\frac{1}{\lambda} J(\theta, \theta_0) = \frac{1}{\lambda n} \sum_{i=1}^{n} \text{Loss}_h(y^{(i)} \cdot (\theta \cdot x^{(i)} + \theta_0)) + \frac{1}{2} |\theta|^2$$

$$= C \sum_{i=1}^{n} \text{Loss}_h(y^{(i)} \cdot (\theta \cdot x^{(i)} + \theta_0)) + \frac{1}{2} |\theta|^2$$



Steps to find c^* (Cross Validation)

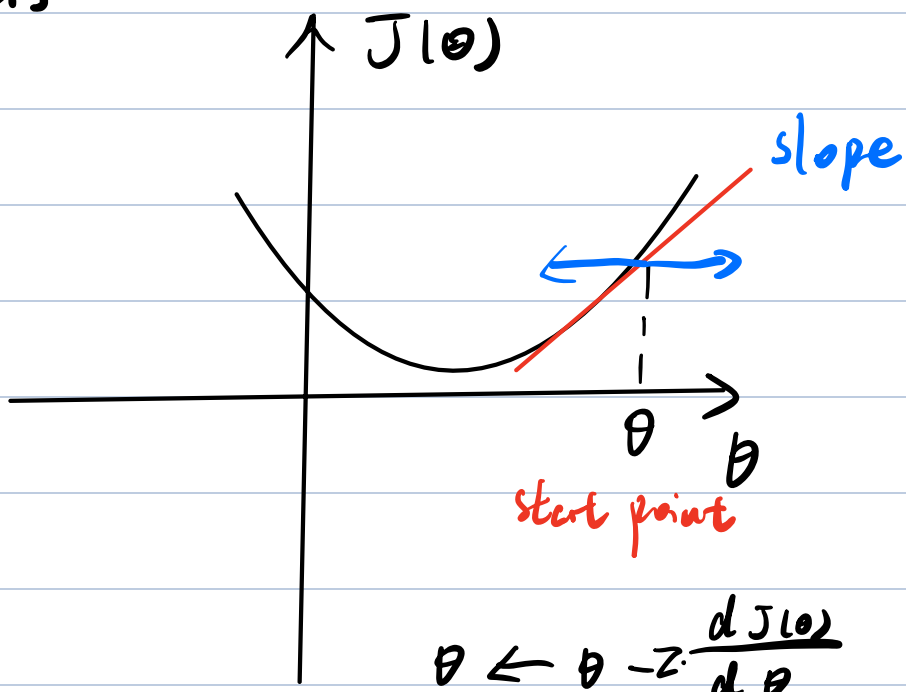
1. divid training sets



2. based on this, find θ and θ_0
3. find C^* based on the new test set

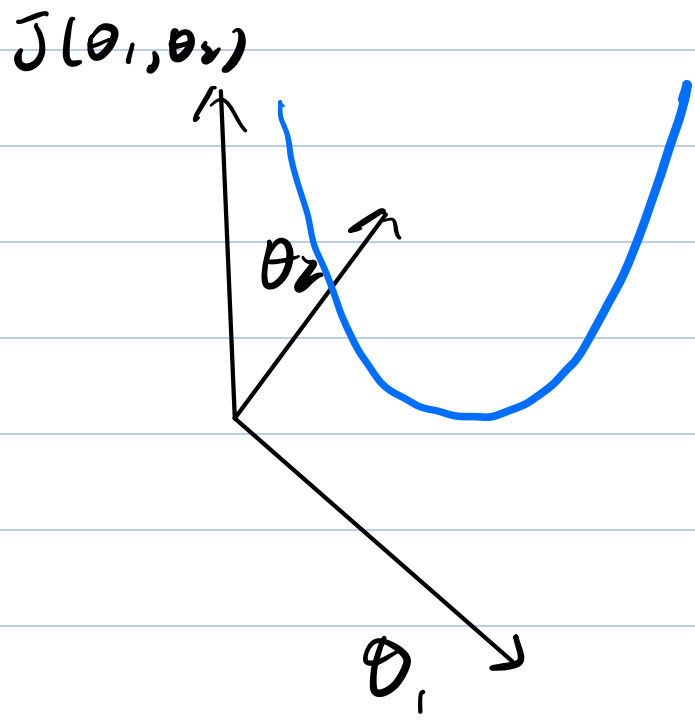
Algorithms to find optimal θ

1. 1 parameters



$$\theta \leftarrow \theta - \underset{\substack{\uparrow \\ \text{learning rate}}}{z} \cdot \frac{dJ(\theta)}{d\theta}$$

2. multi parameters.



$$\theta_j \leftarrow \theta_j - \alpha \cdot \frac{d\, J(\theta_j)}{\theta_j}$$

Stochastic gradient descent

$$J(\theta, \theta_0) = \frac{1}{n} \sum (\text{Loss}(y^{(i)}(\theta \cdot x^{(i)} + \theta_0))) + \frac{\lambda}{2} |\theta|^2$$

$$= \frac{1}{n} \sum \left[\text{Loss}(y^{(i)}(\theta \cdot x^{(i)} + \theta_0)) + \frac{\lambda}{2} |\theta|^2 \right]$$

$$= \frac{1}{n} \sum_{i=1}^{n} J_i(\theta, \theta_0)$$

Sample i at random from $\{1, \ldots, n\}$

$$\theta \leftarrow \theta - Z_t \nabla J_i(\theta)$$

$$\left(Z_t \to 0, \ \sum_{t=1}^{\infty} Z_t = \infty, \ \sum_{t=1}^{\infty} Z_t^2 < \infty \right.$$